1. A two-way set associative cache memory uses blocks of four words. The cache can accommodate a total of 2048 words from main memory. The main memory size is 128K x 32.
   a. How many bits are there in the tag, block and word fields of the address format?
   b. What is the size of the cache memory?

Solution-

128 K = 2^{17}; For a set size of 2, the index: address has 10 bits to accommodate 2048/2 = 1024 words of cache.

(a)

7 bits	10 bits
TAG	INDEX

← Block → Words ←
8 bits 2 bits

b)

Tag 1	Data 1	Tag 2	Data 2	0

1024

← 7 → ← 32 bits → ← 7 → ← 32 bits →

Size of cache memory is 1024 x 2 (7 + 32)
= 1024 × 78

2. A digital computer has a memory unit of 64K x 16 and a cache memory of 1K words. The cache uses direct mapping with a block size of four words.
   a. How many bits are there in the tag, block and word fields of the address format?
   b. How many bits are there in the each word of cache, and how are they divided into functions? Include a valid bit.
   c. How many blocks can the cache accommodate?

Solution-

64 K × 16: 16 bit address; 16-bit data.

(a) 6 8 2 = 16 bits address

TAG	BLOCK	WRD

INDEX = 10 bit cache address.

b)1 6 16 = 23 bits in each word of cache

V	TAG	DATA

(b) 2^8 = 256 blocks of 4 words each

3. Consider a 64KB direct-mapped cache with a line length of 32 bytes.

a. Determine the number of bits in the address that refer to the byte within a cache line.

b. Determine the number of bits in the address required to select the cache line.

Solution-

a. For the given cache, the number of bits in the address to determine the byte within the line=

$n = 2^5$ = 5 bits

b. There are 64K/32= 2048 lines in the given cache. The number of bits required to select the required line 2^{11} = 11 bits

4. Consider a direct mapped cache with 8 cache blocks (0-7). If the memory block requests are in the order-

3, 5, 2, 8, 0, 6, 3, 9, 16, 20, 17, 25, 18, 30, 24, 2, 63, 5, 82, 17, 24

Which of the following memory blocks will be in the cache at the end of the sequence?

Also, calculate the hit ratio and miss ratio.

We have,

- There are 8 blocks in cache memory numbered from 0 to 7.
- In direct mapping, a particular block of main memory is mapped to a particular line of cache memory.
- The line number is given by-

  Cache line number = Block address modulo Number of lines in cache

Solution-

For the given sequence-

- Requests for memory blocks are generated one by one.
- The line number of the block is calculated using the above relation.
- Then, the block is placed in that particular line.
- If already there exists another block in that line, then it is replaced.

Line-0	~~8~~, ~~0~~, ~~16~~, 24
Line-1	~~9~~, ~~17~~, ~~25~~, 17
Line-2	~~2~~, ~~18~~, ~~2~~, 82
Line-3	3
Line-4	20
Line-5	5
Line-6	~~6~~, 30
Line-7	63

Cache Memory

Blocks requests in order
Calculation of line number

3 % 8 = 3	(Miss)
5 % 8 = 5	(Miss)
2 % 8 = 2	(Miss)
8 % 8 = 0	(Miss)
0 % 8 = 0	(Miss)
6 % 8 = 6	(Miss)
3 % 8 = 3	(Hit)
9 % 8 = 1	(Miss)
16 % 8 = 0	(Miss)
20 % 8 = 4	(Miss)
17 % 8 = 1	(Miss)
25 % 8 = 1	(Miss)
18 % 8 = 2	(Miss)
30 % 8 = 6	(Miss)
24 % 8 = 0	(Miss)
2 % 8 = 2	(Miss)
63 % 8 = 7	(Miss)
5 % 8 = 5	(Hit)
82 % 8 = 2	(Miss)
17 % 8 = 1	(Miss)
24 % 8 = 0	(Hit)

- Hit ratio = 3 / 20
- Miss ratio = 17 / 20

5. Consider a fully associative cache with 8 cache blocks (0-7). The memory block requests are in the order-

4, 3, 25, 8, 19, 6, 25, 8, 16, 35, 45, 22, 8, 3, 16, 25, 7

If LRU replacement policy is used, which cache block will have memory block 7?

Also, calculate the hit ratio and miss ratio.

Solution-

We have,

- There are 8 blocks in cache memory numbered from 0 to 7.
- In fully associative mapping, any block of main memory can be mapped to any line of the cache that is freely available.
- If all the cache lines are already occupied, then a block is replaced in accordance with the replacement policy.

Line-0	~~4~~, 45
Line-1	~~3~~, 22
Line-2	25
Line-3	8
Line-4	~~19~~, 3
Line-5	~~6~~, 7
Line-6	16
Line-7	35

Cache Memory

Thus,

- Line-5 contains the block-7.
- Hit ratio = 5 / 17
- Miss ratio = 12 / 17

6.

A cache is having 60% hit ratio for read operation. Cache access time is 30 ns and main memory access time is 100 ns, 50% operations are read operation.
What will be the average access time for read operation?

Avg access time (for read operation) = Hit ratio * cache access time + miss ratio * (Cache access time + MM access time)
Given, Cache access time is 30 ns and main memory access time is 100 ns.
Hit ratio = 60% so Miss ratio = 40%
so Avg access time = (0.6 * 30) + 0.4 * (30 + 100)
= 18 + 52
= 70
so average access time for read operation is 70 ns.

7. In a two-level cache system, the access times of L_1 and L_2 1 and 8 clock cycles, respectively. The miss penalty from the L_2 cache to main memory is 18 clock cycles. The miss rate of L_1 cache is twice that of L_2. The average memory access time(AMAT) of this cache system is 2 cycles. Find the miss rates of L_1 and L_2 .

Access time of $L_1 = T_1 = 1$ clock cycle

Access time of $L_2 = T_2 = 8$ clock cycle

L_2 miss penalty $= T_3 = 18$ clock cycle

L_1 miss rate $= 2 \times L_2$ miss rate

Average memory access time = AMAT = 2 clock cycle

Let, L_2 miss rate = a

$\therefore L_1$ miss rate = 2a

Formula:

$AMAT = (T_1 + 2a \times T_2 + 2a \times a \times T_3)$

Calculation:

$2 = 1 + 2a \times 8 + 2a \times a \times 18$

miss rates of L_1 0.111 and L_2 is 0.056

Average Memory Access time = Hit time + Miss Rate x Miss penalty

Consider a system with 2 level caches. Access times of Level 1 cache, Level 2 cache and main memory are 1 ns, 10ns, and 500 ns, respectively. The hit rates of Level 1 and Level 2 caches are 0.8 and 0.9, respectively. What is the average access time of the system ignoring the search time within the cache?

Average access time = $[H_1 * T_1] + [(1 - H_1) * H_2 * T_2] + [(1 - H_1)(1 - H_2) * H_m * T_m]$
where,

H_1 = Hit rate of level 1 cache = 0.8
T_1 = Access time for level 1 cache = 1 ns
H_2 = Hit rate of level 2 cache = 0.9
T_2 = Access time for level 2 cache = 10 ns
H_m = Hit rate of Main Memory = 1
T_m = Access time for Main Memory = 500 ns

So, Average Access Time = (0.8 * 1) + (0.2 * 0.9 * 10) + (0.2 * 0.1 * 1 * 500)

= 0.8 + 1.8 + 10

= 12.6 ns